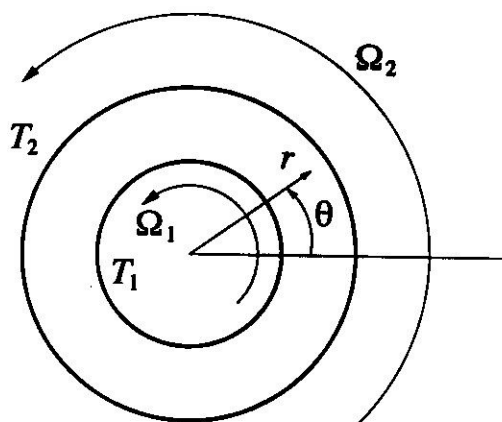


NS2 Consider the steady planar motion of a perfect liquid of density ρ , viscosity μ , thermal conductivity k and specific heat c in the gap formed between two coaxial cylinders of radii R_1 and R_2 that are rotating with angular velocities Ω_1 and Ω_2 , respectively. Because of the symmetry of the problem, $v_z = 0$ and $\partial/\partial z = \partial/\partial\theta = 0$.

1. Obtain the velocity and pressure fields.
2. Calculate the torque exerted on each cylinder, as well as the power needed to keep them rotating.
3. If the temperatures on the cylinder surfaces are T_1 and T_2 , determine the temperature distribution $T(r)$.



1) $\frac{\partial(rv_r)}{\partial r} = 0 \Rightarrow r v_r = \text{const} = 0 \Rightarrow v_r = 0$

θ -mom $\rightarrow 0 = \mu \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) \rightarrow v_\theta = C_1 r + \frac{C_2}{r}$

r -mom $\rightarrow -\rho \frac{v_\theta^2}{r} = -\frac{\partial p}{\partial r} \Rightarrow \frac{\partial p}{\partial r} = \frac{\rho}{r} \left(C_1^2 r^2 + \frac{C_2^2}{r^2} + 2C_1 C_2 \right)$

$v_r(r=R_1) = 0, R_1 \Omega_1$
 $v_r(r=R_2) = 0, R_2 \Omega_2$

$\int_{R_1}^{R_2} \frac{\partial p}{\partial r} r dr = p - p(R_1) = \rho \left[\frac{C_1^2}{2} (r^2 - R_1^2) - \frac{1}{2} C_2^2 \left(\frac{1}{r^2} - \frac{1}{R_1^2} \right) + 2C_1 C_2 \ln\left(\frac{r}{R_1}\right) \right]$

2) $\vec{M}_1 = - \int \vec{r} \wedge \vec{p} \cdot \vec{e}_r R_1 d\theta + \int_0^{2\pi} \vec{r} \wedge \vec{e}_1 \cdot \vec{n} R_1 d\theta = -4\pi\mu C_2 \vec{e}_2 = -4\pi\mu \frac{(R_1 - R_2) R_1^2 R_2^2}{R_2^2 - R_1^2} \vec{e}_2$

$(\tau_{rr}, \tau_{r\theta}, \tau_{rz}) = \mu r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) \vec{e}_\theta = -2\mu \frac{C_2}{R_1^2} \vec{e}_\theta$

$\vec{M}_2 = -\vec{M}_1 = 4\pi\mu \frac{(R_1 - R_2) R_1^2 R_2^2}{R_2^2 - R_1^2} \vec{e}_2$

NOTE: TRY TO APPLY ANGULAR MOMENTUM CONSERVATION TO THE LIQUID

$\dot{W}_1 = \pm \Omega_1 4\pi\mu C_2$
 $\dot{W}_2 = -\Omega_2 4\pi\mu C_2$

3) $0 = \phi_r + \frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \rightarrow T = -\frac{\mu C_2^2}{k r^2} + B_1 \ln r + B_2$

$\phi_r = \mu \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) \right]^2 = \frac{4\mu C_2^2}{r^4}$

$T(r=R_1) = T_1$
 $T(r=R_2) = T_2$

$T - T_1 = \frac{\mu C_2^2}{k} \left(\frac{1}{R_1^2} - \frac{1}{r^2} \right) + \frac{T_2 - T_1 - \frac{\mu C_2^2}{k} \left(\frac{1}{R_1^2} - \frac{1}{R_2^2} \right)}{\ln(R_2/R_1)} \ln\left(\frac{r}{R_1}\right)$

NOTE THAT $\int_0^{2\pi} d\theta \int_{R_1}^{R_2} \phi_r r dr = 2\pi 2\mu C_2^2 \left(\frac{1}{R_1^2} - \frac{1}{R_2^2} \right) = 4\pi\mu C_2 (R_1 - R_2) = \dot{W}_1 + \dot{W}_2 = 2\pi k R_1 \frac{\partial T}{\partial r} \Big|_{R_1} - 2\pi k R_2 \frac{\partial T}{\partial r} \Big|_{R_2}$